

CHRONO EXPRESS

THE LAST HEIST

Game Concept Document · COMS3006A / COMS3025A

1. Concept Overview

Chrono Express: The Last Heist is a 3D train-heist game set aboard a high-security futuristic train crossing a ruined world. The player begins at the rear of the train and must make their way toward the front, moving through three distinct stages before stealing the experimental Chrono Core and stopping the train.

The train is the central structure of the game. Each stage changes the player's objective, the visual identity and the type of challenge, while the story naturally connects them. The result is a progression from a controlled stealth heist, to a time-manipulation robbery, to a chaotic escape through a train that is physically and temporally falling apart.

Core objective: **Board the train → reach the front → steal the Chrono Core → survive the time collapse → stop the train.**

2. Game Structure

The three levels are deliberately designed so that they cannot be described as the same level with different layouts or difficulty. Each introduces a new mechanic and a new gameplay experience.

Level	Location	Experience	Unique element
1 — The Boarding	Station / rear of train	Stealth & infiltration	Guards, cameras, lasers, suspicion and boarding before departure.
2 — The Moving Heist	Moving train	Traversal, puzzles & heist	Time manipulation: slow, freeze, rewind and time ghost.
3 — The Timewreck	Breaking / distorted train	Survival & escape	Unstable time zones, collapsing carriages and final global freeze.

3. Level 1 — The Boarding

Theme: Heist / Stealth

The Chrono Express is preparing to leave a heavily guarded station. The player starts outside the station and must infiltrate the platform and board the train before it departs.

Gameplay

- Sneak past guards and their patrol routes.
- Avoid rotating security cameras.
- Disable or bypass laser barriers.
- Find routes through locked station areas.
- Use a suspicion/alarm system to create a clear fail state.
- Reach the train and override the boarding/train controls.

Visual identity: A luxury railway station with warm lighting, wood, brass, glass, polished surfaces and a controlled night-time atmosphere. The train should be visible throughout enough of the level to establish the destination.

Purpose in progression: This level teaches the player how the world behaves. Cameras rotate, doors close, guards patrol and machinery follows predictable cycles. These predictable behaviours become important once time manipulation is introduced.

Transition: The player boards, the doors close, the station disappears behind them and the Chrono Express begins moving.

4. Level 2 — The Moving Heist

Theme: Train traversal / Time manipulation / Puzzle

The train is now travelling at high speed. The player must move carriage-by-carriage from the rear toward the heavily protected Chrono Core. The train itself becomes part of the gameplay rather than simply being the setting.

Suggested progression: Passenger Carriage → Security Carriage → Cargo Carriage → Mechanical Carriage → Vault.

New mechanic — Time manipulation

After obtaining access to the Chrono technology, the player can temporarily manipulate objects and events in time.

- **Slow:** slows moving guards, machinery, doors and hazards while the player remains mobile.
- **Freeze:** stops a selected object, allowing it to become a temporary platform or safe opening.
- **Rewind:** returns selected objects to an earlier state, such as restoring a collapsed bridge.
- **Time Ghost:** records a previous movement sequence and replays it as a translucent ghost that can hold a switch or pressure plate.

Train-specific set piece: The player can move from inside a carriage onto the roof, where wind, high speed and approaching tunnels create a timing challenge. Slow Time can be used to cross the roof and reach the next carriage safely.

End: The player reaches the vault, breaks in and steals the Chrono Core. The moment it is removed, the train begins experiencing severe temporal instability.

5. Level 3 — The Timewreck

Theme: Escape / Survival / Reality breaking

The familiar train is now the source of the danger. Different sections of the train are experiencing different speeds of time, while the train approaches destroyed track. Stealth is no longer the focus: the objective is to survive and reach the locomotive.

Environmental challenges

- **Frozen carriage:** debris and objects hang motionless in the air and can be used as temporary platforms.
- **Fast-time carriage:** machinery and hazards move at extreme speed; Slow Time becomes essential.
- **Time-loop carriage:** doors and objects repeatedly open, close, fall and rewind, forcing the player to learn the cycle.
- **Breaking train:** carriages disconnect, windows break and objects are thrown through the environment.

Final sequence: The player reaches the roof as the track ahead is destroyed. The Chrono Core is nearly depleted, leaving one final ability: Freeze Time. Everything stops — rain, debris and the train itself — while the player has a short window to sprint toward the locomotive. Time begins restarting behind them as the train collapses.

The player reaches the emergency brake. Time resumes. The train screeches to a stop with the locomotive hanging partially over the destroyed bridge. **Heist complete.**

6. Visual Direction

The visual identity should evolve with the story. This makes each level feel distinct without requiring three completely unrelated environments.

Level	Look	Lighting / atmosphere
1 — The Boarding	Luxury railway station; wood, brass, carpet, glass	Warm station lights, controlled shadows, clean and believable environment
2 — The Moving Heist	Detailed train interiors; passenger, security and cargo areas	Interior practical lights, passing exterior light, security lighting
3 — The Timewreck	Damaged, distorted versions of the train	Emergency lighting, flickering lights, particles, temporal distortion and darker contrast

7. Graphics & Technical Features

Camera / Viewing

A third-person camera will be the main view, with smooth follow and mouse-controlled looking. A closer or first-person view can be used for specific interactions. Cinematic camera moments can be used during boarding, the vault opening and the final train-stop sequence. A simple minimap could show progress toward the locomotive.

Hierarchical modelling

The train will be treated as a parent object containing carriages and their components. Each carriage can contain furniture, doors, lights and machinery. This gives meaningful parent-child relationships: moving the train moves the carriages; rotating a door moves its handle; rotating machinery moves attached components; and train-wide events can affect all child objects.

Lighting, shadows & materials

Multiple light sources will be used to establish the different atmospheres of each stage. Dynamic shadows can be applied selectively. Materials and textures should distinguish wood, metal, glass, carpet and machinery. Reflections, normal/bump information, particles and environmental effects can be added where performance allows.

Animation

Train doors, security cameras, guards, machinery, rotating mechanisms, environmental movement and the player's time ghost provide opportunities for both object animation and hierarchical transformations.

Physics

Physics can be integrated into the moving-train and Timewreck sections: loose crates can move, debris can be thrown or suspended, and environmental objects can respond to impacts or the train's instability.

8. Custom Shader Concept

The main custom shader will be integrated into the game's time system rather than existing only as a visual decoration.

Temporal distortion / Chrono Field shader: when time manipulation is active, selected objects can receive a procedural distortion or scanning effect. A time uniform controls animation, while game-state values determine intensity. The same effect can evolve from a controlled visual in Level 2 into a stronger, unstable distortion in Level 3.

A second possible shader application is the security laser/grid system. The scan can move over time and change behaviour when security is disabled or the alarm is triggered. This provides a clear example of a custom fragment shader driven by time and gameplay state.

9. Gameplay Systems

- **Player movement:** keyboard movement with mouse look.
- **Interaction:** contextual interaction with doors, terminals, switches and objects.
- **Stealth:** guard patrols, camera detection and suspicion/alarm state.
- **Time system:** object-specific time scaling, freezing and state restoration.
- **Health / fail states:** hazards, detection and environmental failures can return the player to a checkpoint.
- **Progression:** completion of each stage unlocks the next train section.
- **Restart:** restart the current level without refreshing the page.

10. How the Concept Meets the Rubric

Rubric area	Planned implementation
Viewing	Third-person camera, mouse look, camera movement through the train, optional first-person/close view, cinematic sequences and potential minimap.
Control & Playability	Keyboard + mouse controls, clear objectives, stealth fail state, time abilities, physics-based hazards and a final escape objective.
3D Effects	Multiple lights, shadows, textured materials, glass/metal effects, particles, environmental motion and temporal visual effects.
Shaders	Custom temporal-distortion and/or security scanning shader using time and gameplay-state uniforms.
Gameplay & Experience	Strong heist narrative with clear progression: infiltrate → steal → escape. Each stage changes the challenge and atmosphere.
Polish	Consistent visual language, HUD, objectives, checkpoints, restart/menu system, loading screen and performance optimisation.
Innovation	Time manipulation integrated into a moving-train heist, including frozen objects, rewinds, time ghosts and a final global time-freeze set piece.
Trailer	Strong cinematic moments: boarding, roof traversal, time freeze, vault opening, train collapse and final brake sequence.

11. Scope & Asset Strategy

The concept is intentionally structured around reusable train assets. Level 2 can be built from modular carriage components, while Level 3 can reuse those same components with changed lighting, damage states, object behaviour, particle effects and time effects. This keeps the project achievable while still making the final level look substantially different.

The team can focus custom modelling effort on high-visibility assets such as the train exterior, carriage doors, vault, Chrono Core and key machinery. Reusable materials and geometry should be preferred where possible to keep browser performance under control.

12. One-Sentence Level Answers

Level 1: A stealth infiltration where the player learns the security systems and boards the train before it leaves.

Level 2: A moving-train heist where the player uses time manipulation to solve obstacles and reach the Chrono Core.

Level 3: A high-speed survival escape where unstable time transforms the train itself into the final obstacle.

13. Final Experience

The intended experience is a compact, cinematic 3D heist that continuously escalates. The player starts with a believable objective — sneak onto a train — then gains a distinctive ability that changes how the familiar environment works. By the final stage, the same train has become a collapsing time anomaly. The concept therefore gives the project a strong visual identity, a clear story, genuinely distinct levels and multiple opportunities to demonstrate the

graphics techniques required by the course.

Tagline: "One train. One heist. Thirty seconds to outrun time."